

Video Streaming Platform (YouTube–Style)

Front End Engineering – II (25CSE0203) • Project Evaluation – I

Project ID: G15–PID–17 (P–17)

Programme: B.E. CSE (Batch 2025) • 3rd Sem

Max Marks: 30 (5M Presentation + 25M Project & Viva)

Tech Stack: React 19, Vite, React Icons, CSS Flexbox & Grid

PROJECT TEAM (GROUP G15)

★ KANAV CHADHA

Roll No: 2510990649

TEAM LEADER

KISHAN

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TEAM MEMBER

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TEAM MEMBER

Problem Statement & Project Objectives

⚠ The Challenge with Modern Media UIs

Complex Component Structure

Modern platforms like YouTube require harmonious synchronization between navigation, dynamic sidebars, category filters, and high-density video feeds.

Responsive Layout Overhead

Seamlessly transitioning from multi-column desktop grids to tablet and single-column mobile views without layout breaks or content distortion.

Search Friction & Typos

Users frequently misspell search queries. Without interactive "Did you mean?" suggestions and instant feedback, discovery fails.

✓ Our Project Goals (Evaluation – I)

Component-Driven React Architecture

Modular decomposition into clean functional components ([Navbar](#), [Sidebar](#), [CategoryChips](#), [VideoCard](#)).

Interactive State Management

Stateful controls using React [useState](#) for sidebar toggles, category filtering, likes, and channel subscriptions.

YouTube Direct Play & Search Feedback

Rich curated cards linking directly to official YouTube streams, coupled with real-time query filtering and fuzzy correction prompts.

System Architecture & Technology Stack

⚙️ React Core & Build Tool

React 19 (Functional Components)

Utilizes pure functional components, JSX syntax, props passing, and unidirectional data flow.

Vite Build Engine

Lightning-fast Hot Module Replacement (HMR) and optimized ES module bundling for swift development.

React Icons Package

Integrated icons ([IoMenu](#), [IoSearchOutline](#), [MdHome](#), [FaHeart](#)) for pixel-perfect YouTube UI.

🎨 CSS & Responsive Layout

CSS Grid for Video Feeds

Dynamic auto-filling grid with `minmax(300px, 1fr)` ensuring fluid card resizing across any monitor.

Flexbox Navigation & Controls

Precision alignment for search inputs, action buttons, channel avatars, and live badges.

Mobile-First Media Queries

Breakpoints at 1024px, 768px, and 480px dynamically toggling full sidebar, mini drawer, and single column view.

📦 Data & State Layer

Curated Video & Channel Store

Structured dataset ([videoData.js](#)) with titles, durations, high-res thumbnails, categories, and direct YouTube URLs.

Fuzzy Keyword Matcher

Client-side lookup matching search tokens against popular topics and channel names for instant suggestions.

Zero External API Dependencies

Self-contained, reliable, and compliant with professor guidelines for Phase-1 evaluation.

Modular Component Hierarchy & Responsibility

Folder & File Structure

```
Youtube_Project/
├── src/
│   ├── components/
│   │   ├── Navbar.jsx (Search bar, mic, icons, logo)
│   │   ├── Sidebar.jsx (Expanded & collapsed drawer)
│   │   ├── CategoryChips.jsx (Horizontal filter pills)
│   │   ├── VideoCard.jsx (Thumbnail, likes, subscribe)
│   │   └── SearchFeedback.jsx (Did you mean prompt)
│   ├── data/
│   │   └── videoData.js (Curated videos & channels)
│   ├── App.jsx (Master state & filter logic)
│   ├── App.css (Theme, Dark mode, Media queries)
│   └── main.jsx (React DOM entry point)
└── package.json
```

Key Component Responsibilities

<Navbar />

Houses the search input, clear button, animated mic button, notification badges, and sidebar collapse toggle.

<Sidebar />

Renders Main navigation, You/Library, Subscriptions list with live status indicators, and Explore categories.

<CategoryChips />

Horizontal pill selector updating `selectedCategory` state on click.

<VideoCard />

Displays 16:9 thumbnail, duration/LIVE tag, channel avatar, verified icon, like counter, and YouTube launch action.

State Management & React Hooks Utilization

⚡ `useState` Implementation Details

1. Sidebar Toggle State (`isSidebarOpen`)

Controls whether the sidebar renders in expanded mode (240px with labels) or compact collapsed mode (72px icons only).

2. Category Filter State (`selectedCategory`)

Holds currently active filter (e.g. 'All', 'React JS', 'Gaming', 'Music'). Automatically highlights active chip.

3. Search Text State (`searchQuery`)

Tracks real-time user keystrokes in the search bar, driving live grid filtering.

4. Likes & Subscriptions State

Maintains arrays of liked video IDs and subscribed channel names, toggling button UI dynamically.

🚀 `useMemo` Optimization & Search Engine

Efficient Video Filtering

Uses `useMemo` to recalculate the filtered video array only when `selectedCategory` or `searchQuery` changes, preventing unnecessary re-renders.

"Did you mean: ...?" Typo Suggestion Engine

Analyzes non-matching query strings and suggests closest topics (e.g. typing "reac" suggests "React JS"; "game" suggests "Gaming").

Zero-State Fallback Screen

When search yields 0 results, renders a helpful fallback prompt with an instant "Explore All Videos" reset action.

UI/UX Features & Responsive Layout System

YouTube Dark Theme

Authentic Color Palette

Built with true dark mode background (`#0f0f0f`), border highlights (`#303030`), and vibrant red brand accents.

Zero Layout Shift (CLS)

All thumbnail containers enforce strict `aspect-ratio: 16 / 9` preventing jumpy image loads.

Multi-Device Responsiveness

Desktop (>1024px)

Full 3–4 column grid with persistent 240px sidebar and full search bar.

Tablet (768px – 1024px)

Adaptive 2–column grid, compact search input, and collapsed icon sidebar.

Mobile (<768px)

Single column feed, hidden collapsed sidebar for maximum media width.

Interactive Micro-Actions

Interactive Like & Subscribe

Instant visual feedback on Like heart icon and Subscribe/Subscribed button states.

Hover Zoom & Play Overlay

Card hover smoothly scales thumbnail and reveals "Watch on YouTube" call-to-action.

Git Workflow & Repository Management

GitHub Best Practices (Rubric 6–8 Marks)

Feature-Based Branching Model

Development organized into clean feature branches (`main`, `feat/navbar-search`, `feat/sidebar-nav`, `feat/video-grid`).

Conventional Commit Messages

Every commit follows semantic prefixes (`feat:`, `fix:`, `style:`, `docs:`) with detailed changelogs.

Clean Repository Structure

Includes complete `.gitignore` (ignoring `node_modules` & `dist`), and comprehensive `README.md`.

Team Task Distribution (Group G15)

★ Kanav Chadha (Team Leader – 2510990649)

Project architecture, Master state management in `App.jsx`, search filtering & Did You Mean algorithm.

● Kishan (2510991457)

Custom `VideoCard` component, Interactive Like/Subscribe state, card hover effects & YouTube deep linking.

● Arsh Behl (2510991354)

Top Navbar design, live search submission, clear buttons, animated voice search mic trigger.

● Soumajit Majee (2510990383)

Responsive Sidebar (expanded/collapsed modes), category chips scrolling bar & media query breakpoints.

Evaluation–I Rubric & Viva Voce Defense (30 Marks)

Rubric Category	Marks	Implemented Project Evidence	Viva Voce Readiness Strategy
Presentation & Communication	5 M	Clear 10–slide deck, role breakdown, architecture diagrams & live demo plan.	Explaining project workflow, individual member roles, and feature roadmap.
HTML Structure & CSS Styling	7–8 M	Semantic HTML5 tags (<code><header></code> , <code><aside></code> , <code><main></code>), CSS Grid & Flexbox, 100% responsive.	Demonstrating media query breakpoints, CSS variables & aspect–ratio properties.
JavaScript & GitHub Basics	6–8 M	ES6+ modules, array filtering (<code>.filter()</code> , <code>.map()</code>), destructuring, Git commits.	Explaining array methods, state immutability, and Git branching history.
DOM & React Basics	6–9 M	Functional components, JSX, Props, <code>useState</code> , <code>useMemo</code> , conditional rendering.	Live coding modification (e.g. adding a new category or state toggle on the spot).

Future Roadmap (Subsequent Evaluations)

Phase 2: Custom Player Engine

HTML5 Media APIs + useRef

Build fully custom in-app video player with scrubber, speed changer, and buffering bar.

Full Keyboard Shortcuts

Space/K (play/pause), J/L (seek 10s), M (mute), F (fullscreen), T (theatre mode).

Phase 3: Watch Page & Comments

Recursive Nested Comments

Multi-level reply tree with upvoting, sorting, and memoized rendering.

Related Videos & Autoplay

Up-next recommendation queue with autoplay countdown timer.

Phase 4: Advanced Capstone

DOM Virtualization

Virtualizing large catalogs with TanStack Virtual for 60 FPS scrolling.

Live Chat & API Integration

Real-time WebSocket chat and backend video ingestion.

EVALUATION READY • GROUP G15

Thank You!

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Members: Kishan (2510991457) • Arsh Behl (2510991354) • Soumajit Majee (2510990383)

✓ Ready for Live Code Demonstration, UI Responsive Testing & Viva Voce Q&A

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